

PETROBOTS MAKER FAIR COMPETITION BOOKLET

2026

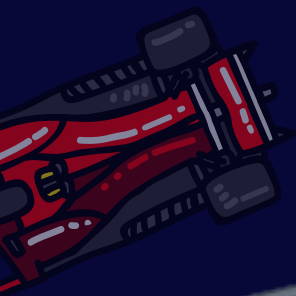
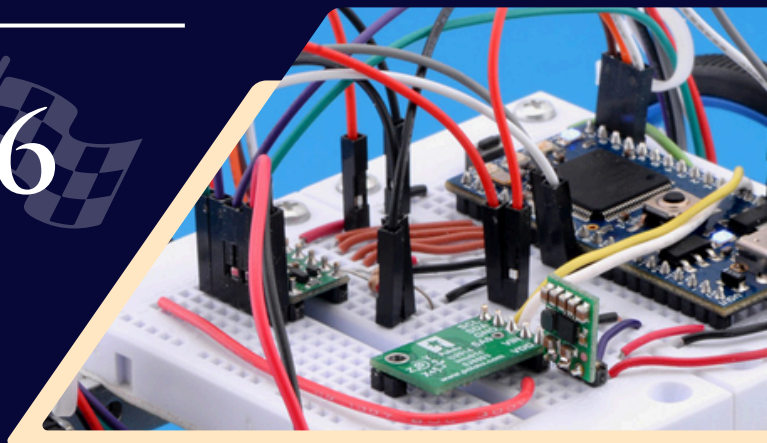


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UTP
UNIVERSITI TEKNOLOGI PETRONAS

SA
STUDENT AFFAIRS



PETROBOTS
UTP ROBOTICS SOCIETY

RUTS

RAJAWANGSALA
UNIVERSITY OF TECHNOLOGY
SEREMBANG



COMPETITION OVERVIEW

PETROBOTS ROBOTRACK GRAND PRIX (ROBOTRACK GP)

PETROBOTS RoboTrack Grand Prix (RoboTrack GP) is an **international line-following robot competition** will be organized as the main competitive activity of PETROBOTS Maker Fair 2026. Participants will **design and program autonomous robots capable of completing line-following challenges** based on surprise maps or tasks revealed during the competition. The event aims to strengthen students' understanding of robotics fundamentals such as sensor integration, control logic, and embedded programming, while encouraging problem-solving, teamwork, and hands-on engineering experience. This competition also provides participants with an opportunity to apply theoretical knowledge in a practical setting and will be open to participants from local and invited international institutions.



OBJECTIVES



1. TO STRENGTHEN STUDENTS' UNDERSTANDING OF ROBOTICS FUNDAMENTALS

This competition aims to deepen participants' knowledge of core robotics concepts such as sensor integration, control algorithms, embedded systems, and autonomous navigation through practical implementation and experimentation.



2. TO ENHANCE PROBLEM-SOLVING AND ANALYTICAL THINKING SKILLS

By introducing surprise track layouts and real-time challenges, participants are encouraged to analyze situations critically, optimize their robot performance, and adapt their programming strategies under time constraints.



3. TO PROMOTE TEAMWORK, COLLABORATION, AND TECHNICAL COMMUNICATION

Participants will work in teams to design, build, and troubleshoot their robots, fostering effective communication, task delegation, leadership skills, and collaborative engineering practices.



4. TO PROVIDE A PLATFORM FOR APPLYING THEORETICAL KNOWLEDGE IN A REAL-WORLD CONTEXT

The competition offers a hands-on learning experience where students can transform classroom concepts into functional robotic systems, bridging the gap between academic learning and practical engineering application.



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RUTS

RAHMANULLAH
UNIVERSITY
OF KUALA LUMPUR
SRIVIJAYA



ELIGIBILITY & ENTRY FEES

1. ELIGIBILITY

- The competition is open to registered student teams from local and invited international institutions.
- Each team shall consist of one **(1)** to three **(3)** members.
- Each team is permitted to register one **(1)** robot only.
- All participants must complete registration within the deadline.
- The competition is open to **local and international student participants.**
- Participants must register under one of the following categories:
 - **Expert Category**
 - **Grandmaster Category**
- Each team must complete the official registration process before the deadline.
- Registration is only confirmed upon successful payment of the entry fee.

ELIGIBILITY & ENTRY FEES

2. ENTRY FEE STRUCTURE

2.1 LOCAL PARTICIPATION (MALAYSIA)

CATEGORY	ENTRY FEE (PER TEAM)
EXPERT CATEGORY	RM200.00
GRANDMASTER CATEGORY	RM300.00

2.2 INTERNATIONAL PARTICIPATION

CATEGORY	ENTRY FEE (PER TEAM)
EXPERT CATEGORY	100.00 USD
GRANDMASTER CATEGORY	150.00 USD

ELIGIBILITY & ENTRY FEES

3. REGISTRATION FEE PACKAGES

3.1 LOCAL PARTICIPATION INCLUDES:

- Participation Certificate
- Door Gifts
- Food and Beverage (3 meals per day)
- Access to The Podium Lounge (Award Night)

3.2 INTERNATIONAL PARTICIPATION INCLUDES:

- Participation Certificate
- Door Gifts
- Transportation
- Food and Beverage (3 meals per day)
- Access to The Podium Lounge (Award Night)
- Accommodation (Optional)

(Eligible for East Malaysia and International teams only)

- City Tour: Ipoh / Kuala Lumpur / Penang (Optional)

ELIGIBILITY & ENTRY FEES

4. ADDITIONAL OPTIONAL FEES

4.1 CITY TOUR PACKAGES

CITY TOUR (IPOH)	RM 100
CITY TOUR (KUALA LUMPUR)	RM 100
CITY TOUR (PENANG)	RM 100

5. ACCOMMODATION FEES (PER PAX PER NIGHT)

HOTEL / UTP CAMPUS	PRICE (RM)
UTP HOSTEL (WITHOUT AIRCOND)	RM30
UTP HOSTEL (WITH AIRCOND)	RM60

ELIGIBILITY & ENTRY FEES

6. PRIZES

EXPERT CATEGORY

ROBOTRACK GP <u>EXPERT</u> 1ST PRIZE	RM 2000
ROBOTRACK GP <u>EXPERT</u> 2ND PRIZE	RM 1500
ROBOTRACK GP <u>EXPERT</u> 3RD PRIZE	RM 1000

GRANDMASTER CATEGORY

ROBOTRACK GP <u>GRANDMASTER</u> 1ST PRIZE	RM 3000
ROBOTRACK GP <u>GRANDMASTER</u> 2ND PRIZE	RM 2000
ROBOTRACK GP <u>GRANDMASTER</u> 3RD PRIZE	RM 1500

COMPETITION GUIDELINES



COMPETITION FORMAT

1. The competition shall consist of preliminary and final rounds.
2. The competition track layout will be revealed on the day of the event.
3. Each team will be granted a designated preparation period before the official run.
4. Each team is allowed two (2) official attempts per round.
5. The best recorded completion time will be taken as the final score.



ROBOT REQUIREMENTS

1. The robot must operate fully autonomously throughout the run.
2. Remote control, wired control, Bluetooth, Wi-Fi, or any external communication is strictly prohibited during operation.
3. The robot must comply with the specified dimension and weight limitations set by the organizer.
4. All power sources must be onboard and securely mounted.
5. The robot must not damage or modify the competition track in any way.



TRACK SPECIFICATIONS

1. The track consists of a contrasting line on a flat surface.
2. The track may include straight paths, curves, sharp turns, intersections, and designated checkpoints.
3. Track dimensions and line width will be announced prior to the competition.
4. The robot must begin from a stationary position at the designated starting line.

COMPETITION GUIDELINES



SCORING AND PENALTIES

1. The winner shall be determined based on the fastest completion time.
2. Time penalties may be imposed for:
 - **Leaving the track**
 - **Manual intervention**
 - **Failure to follow track requirements**
3. Failure to complete the course within the maximum time limit may result in disqualification.
4. In the event of a tie, an additional run may be conducted.



TECHNICAL INSPECTION

1. All robots must undergo technical inspection prior to the competition.
2. Robots that fail to meet specifications must be modified before participating.
3. The organizing committee reserves the right to disqualify any robot that violates the rules.

PARTICIPANTS' GUIDELINES



CODE OF CONDUCT

1. Participants must demonstrate professionalism, integrity, and sportsmanship at all times.
2. Any form of interference with other teams is strictly prohibited.
3. Participants must comply with all instructions issued by judges and organizers.
4. The decision of the judges shall be final and binding.



RESPONSIBILITIES OF PARTICIPANTS

1. Teams are responsible for the safety and security of their equipment.
2. Participants must attend all briefings and adhere to the official competition schedule.
3. Teams must ensure their robot complies with all technical and safety regulations prior to inspection.



SAFETY AND COMPLIANCE

1. All electrical wiring must be properly insulated and securely mounted.
2. Hazardous materials or unsafe components are strictly prohibited.
3. Participants must adhere to venue safety rules at all times.
4. Any unsafe behavior may result in immediate disqualification

RULES & REGULATION

1.0 TOURNAMENT OVERVIEW

The **PETROBOTS RoboTrack Grand Prix (RoboTrack GP)** is an official tournament designed to showcase the intersection of mechanical engineering, electronic integration, and algorithmic programming. This tournament provides a competitive yet collaborative environment where students and hobbyists can apply theoretical STEM knowledge to solve real-world autonomous navigation challenges. By establishing three distinct categories which are Rookie, Expert, and Grandmaster. This tournament ensures that participants are evaluated against peers of similar technical experience and hardware capabilities. All attendees, including participants, mentors, and spectators, are expected to uphold the principles of fair play and technical honesty while adhering strictly to the regulations outlined in this manual.

2.0 TOURNAMENT OBJECTIVE

The primary objective of **RoboTrack GP** is to challenge participants to design, build, and program a fully autonomous robot capable of navigating a complex path with speed and high precision. Beyond the technical aspect, the event aims to cultivate critical thinking and real-time problem-solving skills as teams calibrate their sensors to adapt to the specific environment and lighting conditions. Furthermore, the tournament is structured to encourage mastery of various control platforms, from modular educational kits to advanced open-source microcontrollers. Ultimately, the goal is to foster innovation and engineering excellence, rewarding teams that can achieve the most efficient balance between hardware stability and sophisticated software logic.

RULES & REGULATION

3.0 CATEGORY CLASSIFICATION

To ensure equitable competition, the tournament is divided into three distinct categories based on technical experience and hardware capabilities:

A. Expert : Primary/Secondary

B. Grandmaster : Open/University Level

4.0 CATEGORY B: EXPERT

4.1 ELIGIBILITY

- **Education Level:** This category is strictly limited to students currently enrolled in **Primary School** or **Secondary School** (High School) programs.
- **Team Composition:** Teams must consist of a minimum of one (1) and a maximum of three (3) students.

NOTE: One (1) Adult Mentor (Teacher/Parent) is allowed for supervision but is strictly prohibited from programming or handling the robot during the tournament.

RULES & REGULATION

4.2 HARDWARE SPECIFICATIONS

4.2.1 MICROCONTROLLER

- Participants are permitted to use either **LEGO-based platforms** or **Arduino-based microcontrollers**, provided they adhere to the strict component limitations.
- Only official or clone 8-bit Atmel microcontrollers including the Uno, Nano, Mega, or **Pro Mini** are permitted.
- Only **LEGO-branded controllers including Spike Prime, Spike Essential, Mindstorms EV3, or NXT** are permitted.
- Strictly **one (1) hub** per robot. No daisy-chaining or dual-hub configurations.
- Wi-Fi connections must be **turned off** during the run.
- Bluetooth connections will be allowed **only** under the supervision of the organising committees
- **Prohibited Controllers:** ESP32, STM32, Raspberry Pi Pico, Teensy, or any 32-bit ARM-based boards

RULES & REGULATION

4.2.2 MOTORS & DRIVE TRAIN

- Both LEGO and Arduino builds are strictly limited to a maximum of **two (2) motors**, dedicated solely to locomotion.
- **Third motors** for steering, lifting, or decoration are prohibited.
- Motors must be rated for **5V to 6V**.
- **High-speed brushless motors** or **stepper motors** are strictly prohibited.
- Only standard **LEGO-branded motors** compatible with the chosen hub are permitted.
- Rotary encoders for speed/distance measurement are **not allowed** in this category.
- The **maximum allowable motor speed** is **300 RPM**.
- Any standard H-Bridge motor driver is allowed (e.g., L298N, L293D, TB6612FNG).

RULES & REGULATION

4.2.3 SENSORS

- Strictly limited to a maximum of **four (4) Infrared (IR) sensor units** (e.g., TCRT5000, TCRT5000L).
- **Maximum of four (4) LEGO Color or Light sensors** can be connected to the hub.
- If using an integrated sensor array (e.g., QTR-3RC), **a maximum of four (4) total numbers of IR sensors on the integrated sensor array** can be electrically connected to the Arduino.
 - The unused sensors must have their **jumper wires** physically **removed**, or signal paths cut.
 - Covering unused sensors with tape is **not sufficient**; they must be electrically disconnected.
- On-board potentiometers for threshold adjustment are allowed.

4.2.4 POWER SUPPLY & CHASSIS

- **Total battery voltage** must not exceed 9.6V nominal.
 - **Allowed:** 6x AA 9V Batteries, 1x 9V Block, or 2S Li-ion/LiPo (7.4V).
 - **Prohibited:** 3S LiPo (11.1V) or higher.
- Can be **3D printed, acrylic, wood, or metal**. Must be a **"fixed" chassis** (no moving parts during the run except wheels).
- **LEGO builds** must use the standard rechargeable battery pack provided by the manufacturer or standard AA batteries.
- **Dimensions** must fit within a 25 cm x 25 cm footprint.

RULES & REGULATION

4.3 DISQUALIFICATION (DQ) CRITERIA

- Any robot found to have more than **two (2) active sensors** connected to the microcontroller or hub during the run will be disqualified.
- For multi-sensor arrays, failure to physically disconnect all but **two(2) signal wires** will result in an immediate DQ.
- Any **custom-built robot** utilizing a power supply **exceeding 9.6V nominal** (such as a 3S Lithium-Polymer battery) will be barred from competing.
- Any robot with an active **Bluetooth, Wi-Fi, or Radio Frequency (RF) link** during the official run will be disqualified to ensure the robot is operating fully autonomously.
- Using sticky tires that leave residue or metal parts that scratch the track surface.
- If the judges detect the use of **motor encoders, gyroscopes, or IMU sensors** to assist in navigation or speed control, the robot will be disqualified, as navigation must rely solely on the two permitted line sensors.
- The use of a **third motor** for any purpose, or the use of unauthorized motor types such as brushless (BLDC) or stepper motors, is strictly prohibited.

RULES & REGULATION

5.0 CATEGORY C: GRANDMASTER

5.1 ELIGIBILITY

- **Education Level:** This category is opened to University students, Technical College students, and Professionals (Open Age).
- **Team Composition:** Teams must consist of a minimum of one (1) and a maximum of three (3) participants.

5.2 HARDWARE SPECIFICATIONS

5.2.1 MICROCONTROLLER

- There are no restrictions on the type of **microcontroller, sensors, or motors** used in this category.(e.g., ESP32, STM32, Teensy 4.0/4.1, Raspberry Pi Pico, Raspberry Pi 4/5, Jetson Nano).)
- Participants are encouraged to design and use their own custom **printed circuit boards (PCB)**.
- Wireless modules (**Bluetooth/Wi-Fi**) must be **disabled** during the official run.

RULES & REGULATION

5.2.2 MOTORS & DRIVE TRAIN

- No limit on type or quantity.
- High-performance Coreless DC, Brushless DC (BLDC), or Stepper motors are permitted.
- Any material is allowed (Silicone, Foam, Rubber). However, tires must not be coated in "sticky" chemical substances that transfer to and damage the track

5.2.3 SENSORS

- **No limit** on the number of sensors.
- Open to **IR Arrays** (e.g., QTR-8HD), **CCD/CMOS Line Cameras**, **Laser Time-of-Flight (ToF) sensors**, and **Ultrasonic sensors**.
- **Encoders** are highly recommended for **high-speed velocity control and odometry**

5.2.4 POWER SUPPLY & CHASSIS

- Maximum voltage of **12.6V nominal** (e.g., 3S LiPo battery).
- All **LiPo batteries** must be securely mounted and show no signs of swelling or damage.
- **Dimensions:** Must fit within a 25 cm x 25 cm footprint at the start.

RULES & REGULATION

5.3 DISQUALIFICATION (DQ) CRITERIA

- Use of a power supply **exceeding 12.6V nominal** (e.g., using a 4S Li-Po battery or higher), which poses a safety risk to the track and equipment.
- Any robot with an active **Bluetooth, Wi-Fi, or Radio Frequency (RF) link** during the official run will be disqualified to ensure the robot is operating fully autonomously.
- Use of **metal components, sharp chassis edges, or high-torque motors** that scratch, gouge, or otherwise damage the tournament surface.
- Application of **"sticky" chemical compounds, glues, or liquids** to the tires that leave a residue on the black line or white surface.
- Use of exposed, damaged, or "puffed" **Li-Po batteries**, or robots with high-current wiring that lacks proper insulation or a master kill-switch.

RULES & REGULATION

6.0 UNIVERSAL GAMEPLAY AND SCORING

The following **gameplay procedures** and **scoring metrics** apply to all three tournament categories to ensure a standardized evaluation process.

6.1 THE OFFICIAL RUN

Each participating team is granted **two (2) official attempts** to navigate the course. The official timer starts the moment the leading edge of the robot crosses the start line and stops when the robot successfully crosses the finish line. Before the official heat begins, each team is allowed a three-minute calibration window on the track to adjust their sensors to the ambient lighting and surface conditions.

- The robot must start behind the "**Start Line**" and fit within a 25 cm x 25 cm footprint.
- One team member presses the "**Run**" button. No other touch is allowed.
- The robot must follow a **19 mm-width black line**.
- If the robot deviates entirely from the line (no part of the robot is over the line), it is considered a **Failure**.

RULES & REGULATION

6.2 CHECKPOINT AND REVIVAL SYSTEM

To ensure that all teams have the opportunity to complete the course, a **checkpoint system** is implemented. A checkpoint is cleared only when the entire body of the robot has crossed the marker. If a robot loses the line, remains stationary for more than **five (5) seconds**, or moves in the wrong direction, it is considered a **failure**, a team member may pick up the robot and place it back at a position behind the last successfully cleared checkpoint. The timer will not be paused during this **revival process**.

- **Checkpoints:** Marked by horizontal blue lines.
- The robot has "**passed**" once its **entire body** (rear bumper) clears the checkpoint line.
- **Revival Procedure:**
 1. The Judge calls "**Line Out.**"
 2. The Team Member picks up the robot.
 3. The robot is placed **completely behind** the last successfully passed checkpoint.
 4. The timer **does not stop** during this process.
- Each time the robot is touched for a revival, a **+10 second** penalty is added.

RULES & REGULATION

6.3 SCORING AND PENALTIES

The final ranking is determined by the **Total Corrected** Time. This time is calculated by taking the raw run time and adding a **ten-second** penalty for every manual touch or revival performed during the run. In the event of a tie in the final corrected time, the team with the fewer number of revivals will be declared the winner.

- The winner is the team with the lowest **Final Score (S)**:

$$S = T + (N \times 10)$$

- T: **Total time** from Start to Finish.
- N: Number of **Revivals**.

7.0 TECHNICAL INSPECTION AND SCRUTINEERING

- Every robot must undergo a **mandatory technical inspection** by the judges 30 minutes before the tournament begins.
- Robots that **do not meet** the hardware specifications for their category will be given **15 minutes** to make corrections.
- Any robot that **fails** the **second inspection** will be moved to a **higher category** or **disqualified** at the judges' discretion.
- Once a robot has passed inspection, it must **remain** in the designated "Parc Fermé" area and cannot be modified until its run.

RULES & REGULATION

8.0 DISQUALIFICATION AND PENALTIES

- Any robot that causes **physical damage** to the track surface or the timing equipment will be immediately disqualified.
- The use of **remote controls, Bluetooth, or any wireless communication** to influence the robot during a run is strictly prohibited.
- Applying **sticky substances** or **chemical liquids** to the tires to increase traction is not allowed and will result in a disqualification.
- Teams that demonstrate **poor sportsmanship** or **harass** the judges and other competitors will be asked to leave the venue.

9.0 APPEALS AND DISPUTES

- All **results announced** by the finish line judges are considered **provisional** until verified by the Head Judge.
- Any **protest** regarding a run must be filed by the Team Captain to the Head Judge within 5 minutes of the run's completion.
- Protests must be based on a **specific rule violation** and not on a **disagreement** with a judge's subjective timing.
- The Head Judge may **review video evidence** if available, but their final decision is **absolute** and **cannot be challenged**.

RULES & REGULATION

10.0 SAFETY AND LIABILITY

- Participants are responsible for the **safe handling** of their **batteries**, particularly Lithium-Polymer (LiPo) packs.
- Any robot that begins to smoke or poses a **fire hazard** will be **removed** from the track immediately by the safety marshals.
- The organizers are **not responsible** for **any damage** to robots or personal property occurring during the tournament.
- These **rules** are subject to change by the **organizing committee**, provided that all teams are notified before the start of the event.

11.0 TRACK AND ENVIRONMENT SPECIFICATIONS

- The **tournament track** is constructed from a matte white material to minimize reflections and glare.
- The **tracking path** is marked with a black line consisting of non-reflective material with a width of 18 mm - 20 mm.
- **Lighting conditions** at the venue are provided by standard indoor fixtures; however, participants must be prepared for minor variations in light intensity.
- **Flash photography** by spectators is strictly **prohibited** near the track as it may interfere with the robot's infrared sensors.
- The **minimum turn radius** for the rookie and expert tracks is 15 cm, while the grandmaster track may include sharper turns.
- Surfaces are generally level, but participants should account for a **height tolerance** of 3 mm at track joints.

RULES & REGULATION

12.0 FINAL AUTHORITY AND STATEMENT

- The **organizing committee** reserves the right to **amend** these rules at any time to ensure the safety and fairness of the tournament.
- All participants **agree** to be **photographed or filmed** for promotional purposes by the event organizers.
- By entering the **tournament area**, all team members and mentors agree to follow the **instructions** of the event staff and marshals.
- The **decisions** made by the Head Judge regarding scores, penalties, and winners are **final** and **legally binding**.

EVALUATION & JUDGING MATERIAL

1. JUDGING AUTHORITY

1. The competition will be overseen by an official panel appointed by the organizing committee.
2. The panel includes the Head Judge, who has overall authority, and Timekeepers, responsible for recording accurate times.
3. Judges have full authority to make decisions on performance, rule interpretation, and safety issues.
4. All decisions made by the judges are final and binding, and participants must respect their authority.

2. EVALUATION OVERVIEW

1. Robots are evaluated based on performance, accuracy, technical compliance, and speed.
2. The primary objective is achieving the lowest Final Corrected Time.
3. Both technical compliance and behavior on the track are considered in the evaluation.
4. Judges assess not only speed but also how reliably and correctly the robot completes the course.

3. PERFORMANCE EVALUATION

1. The robot must follow the black line continuously without leaving the track.
2. Smooth and stable navigation through straight paths and turns is required.
3. Checkpoints must be completed successfully with minimal or no manual revivals.
4. The mandatory 3-second stop challenge must be executed correctly.
5. Frequent errors, such as veering off-track or skipping checkpoints, will result in penalties.

EVALUATION & JUDGING MATERIAL

4. SPEED AND TIME MEASUREMENT

1. Timing begins when the robot crosses the Start Line and stops at the Finish Line.
2. Faster completion times are preferred but only if all rules are followed.
3. Judges monitor for shortcuts, off-track behavior, or rule violations that may invalidate raw times.
4. Accurate timing ensures fair competition and proper ranking of teams.

5. PENALTIES AND TIME ADJUSTMENT

1. Each manual revival adds 10 seconds to the total time.
2. Incorrect execution of the stop challenge incurs additional penalties.
3. All penalties are added to the raw run time to calculate the Final Corrected Time.
4. Judges maintain clear records of all penalties to ensure transparency.
5. Repeated rule violations may result in disqualification at the judge's discretion.

6. FINAL SCORE CALCULATION

1. The final score is calculated using the formula:

$$S = T + (N \times 10)$$

- T = Raw time from Start to Finish
 - N = Number of manual revivals
2. The robot with the lowest final score wins.
 3. This formula balances speed, technical compliance, and consistency.
 4. Judges review all calculations for accuracy and fairness.

EVALUATION & JUDGING MATERIAL

7. TIE-BREAKING RULES

1. If two or more teams have the same final score, the team with fewer revivals ranks higher.
2. If still tied, the team with the faster raw completion time ranks higher.
3. If a tie persists, judges may use discretion based on overall performance and technical execution.

8. TECHNICAL COMPLIANCE ASSESSMENT

1. Judges check robots before and during runs for compliance with category rules.
2. Checks include controller type, sensor count, motor count, battery voltage, and wireless communication.
3. Non-compliance may result in penalties or disqualification, even if the robot completes the course.
4. Judges may stop a run immediately if the robot is non-compliant.

9. OBSERVATION DURING THE RUN

1. Judges monitor for unauthorized assistance or external communication.
2. Unsafe behavior or actions that could damage the track are prohibited.
3. Violations result in penalties or immediate disqualification.
4. Safety is a top priority, and judges enforce rules to protect participants and equipment.

10. JUDGE'S FINAL DECISION

1. Judges may stop a run if a robot poses any safety risk.
2. Judges have full authority to interpret situations not clearly covered in the rules.
3. All official results announced by the Head Judge are final.
4. Teams are expected to accept these decisions without dispute to maintain fairness.

TERMS AND CONDITIONS (T&C)

1. JUDGING AUTHORITY

1. The competition will be overseen by an official panel appointed by the organizing committee.
2. The panel includes the Head Judge, who has overall authority, and Timekeepers, responsible for recording accurate times.
3. Judges have full authority to make decisions on performance, rule interpretation, and safety issues.
4. All decisions made by the judges are final and binding, and participants must respect their authority.

2. EVALUATION OVERVIEW

1. Robots are evaluated based on performance, accuracy, technical compliance, and speed.
2. The primary objective is achieving the lowest Final Corrected Time.
3. Both technical compliance and behavior on the track are considered during evaluation.
4. Judges assess not only speed but also how reliably and correctly the robot completes the course.

3. PERFORMANCE EVALUATION

1. The robot must follow the black line continuously without leaving the track.
2. Smooth and stable navigation through straight paths and turns is required.
3. Checkpoints must be completed with minimal to no manual revivals.
4. The mandatory 3-second stop challenge must be executed correctly.
5. Frequent errors (e.g., veering off-track, skipping checkpoints) will result in penalties.

TERMS AND CONDITIONS (T&C)

4. SPEED AND TIME MEASUREMENT

1. Timing begins when the robot crosses the Start Line and stops when it crosses the Finish Line.
2. Faster completion times are preferred but only if all rules are followed.
3. Judges monitor for shortcuts, off-track behavior, or rule violations that may invalidate raw times.
4. Accurate timing ensures fair competition and proper ranking of teams.

5. PENALTIES AND TIME ADJUSTMENT

1. Each manual revival adds 10 seconds to the total time.
2. Incorrect execution of the stop challenge incurs additional penalties.
3. All penalties are added to the raw run time to calculate the Final Corrected Time.
4. Judges maintain clear records of all penalties to ensure transparency.
5. Repeated rule violations may result in disqualification at the judge's discretion.

6. FINAL SCORE CALCULATION

1. The final score is calculated using the formula:

$$S = T + (N \times 10)$$

T = Raw time from Start to Finish

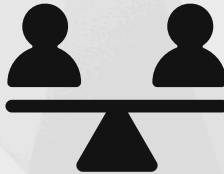
N = Number of manual revivals

2. The robot with the lowest final score wins.
3. This formula balances speed, technical compliance, and consistency.
4. Judges review all calculations for accuracy and fairness.

TERMS AND CONDITIONS (T&C)

7.TIE-BREAKING RULES

- 1.If two or more teams have the same final score, the team with fewer revivals ranks higher.
- 2.If still tied, the team with the faster raw completion time ranks higher.
- 3.If a tie persists, judges may use discretion based on overall performance and technical execution.



8.technical compliance assessment

- 1.Judges check robots before and during runs for compliance with category rules.
- 2.Checks include controller type, sensor count, motor count, battery voltage, and wireless communication.
- 3.Non-compliance may result in penalties or disqualification, even if the robot completes the course.
- 4.Judges may stop a run immediately if the robot is non-compliant.

TERMS AND CONDITIONS (T&C)

9.OBSERVATION DURING THE RUN

- 1.Judges monitor for unauthorized assistance or external communication.
- 2.Unsafe behavior or actions that could damage the track are prohibited.
- 3.Violations result in penalties or immediate disqualification.
- 4.Safety is a top priority, and judges enforce rules to protect participants and equipment.

10.JUDGE'S FINAL DECISION

- 1.Judges may stop a run if a robot poses any safety risk.
- 2.Judges have full authority to interpret situations not clearly covered in the rules.
- 3.All official results announced by the Head Judge are final.
- 4.Teams are expected to accept these decisions without dispute to maintain fairness.

EVENT CONTACT DETAILS

For any inquiries regarding the PETROBOTS RoboTrack Grand Prix, participants may contact the organizing committee through the following official representatives



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All official inquiries, clarifications, and competition-related matters should be directed to the contacts listed above. Participants are advised to communicate professionally and clearly state their team name and category when making inquiries.



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